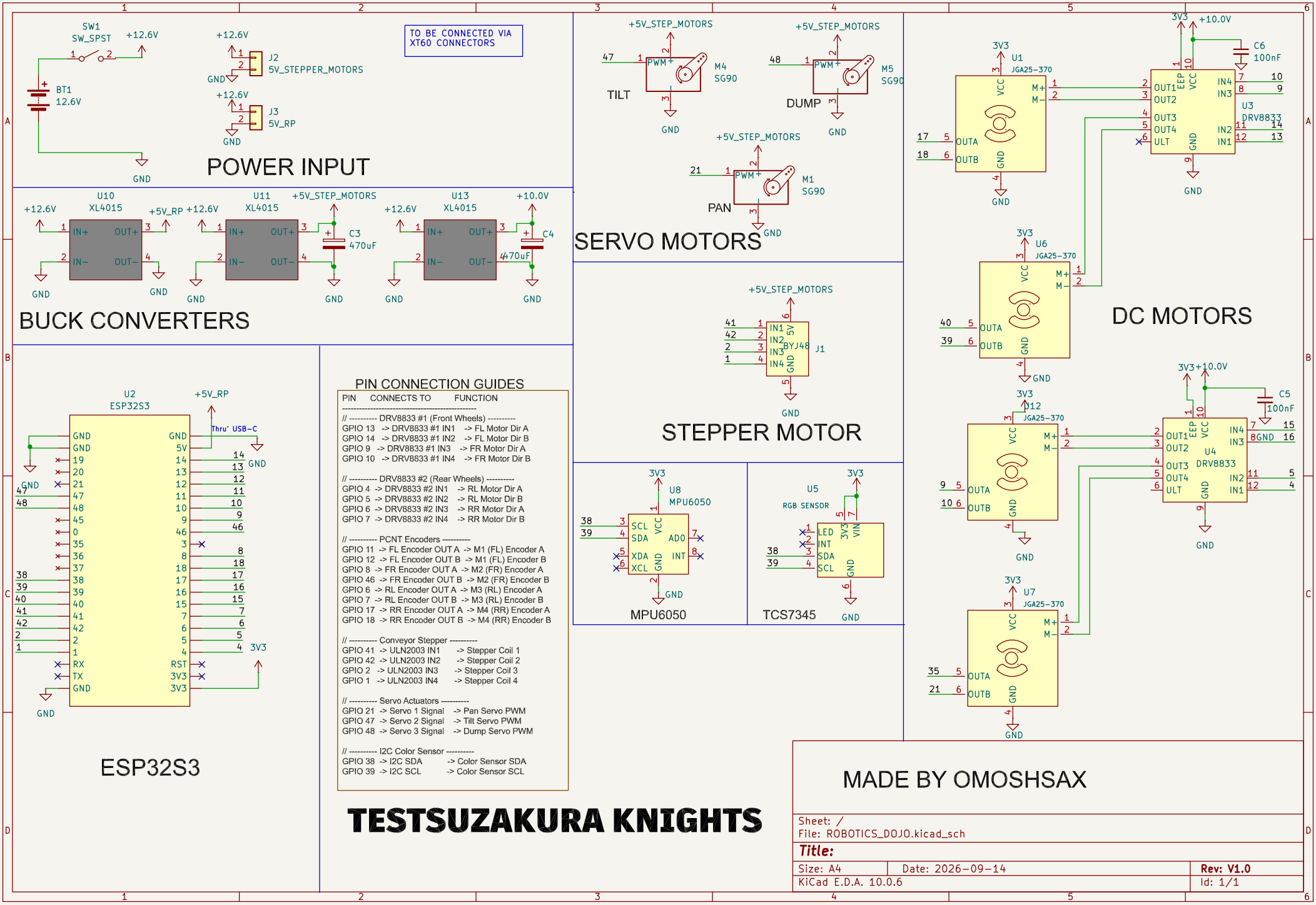

SW1
SW_SPST
+12.6V
BT1
12.6V
GND
TO BE CONNECTED VIA XT60 CONNECTORS
J2
5V_STEPPER_MOTORS
J3
5V_RP
POWER INPUT
U10
XL4015
U11
XL4015
U13
XL4015
+5V_RP
+5V_STEP_MOTORS
+10.0V
C3
470uF
C4
470uF
BUCK CONVERTERS
U2
ESP32S3
Thru' USB-C
ESP32S3
PIN CONNECTION GUIDES
PIN CONNECTS TO FUNCTION
// ---------- DRV8833 #1 (Front Wheels) ----------
GPIO 13 -> DRV8833 #1 IN1 -> FL Motor Dir A
GPIO 14 -> DRV8833 #1 IN2 -> FL Motor Dir B
GPIO 9 -> DRV8833 #1 IN3 -> FR Motor Dir A
GPIO 10 -> DRV8833 #1 IN4 -> FR Motor Dir B
// ---------- DRV8833 #2 (Rear Wheels) ----------
GPIO 4 -> DRV8833 #2 IN1 -> RL Motor Dir A
GPIO 5 -> DRV8833 #2 IN2 -> RL Motor Dir B
GPIO 6 -> DRV8833 #2 IN3 -> RR Motor Dir A
GPIO 7 -> DRV8833 #2 IN4 -> RR Motor Dir B
// ---------- PCNT Encoders ----------
GPIO 11 -> FL Encoder OUT A -> M1 (FL) Encoder A
GPIO 12 -> FL Encoder OUT B -> M1 (FL) Encoder B
GPIO 8 -> FR Encoder OUT A -> M2 (FR) Encoder A
GPIO 46 -> FR Encoder OUT B -> M2 (FR) Encoder B
GPIO 6 -> RL Encoder OUT A -> M3 (RL) Encoder A
GPIO 7 -> RL Encoder OUT B -> M3 (RL) Encoder B
GPIO 17 -> RR Encoder OUT A -> M4 (RR) Encoder A
GPIO 18 -> RR Encoder OUT B -> M4 (RR) Encoder B
// ---------- Conveyor Stepper ----------
GPIO 41 -> ULN2003 IN1 -> Stepper Coil 1
GPIO 42 -> ULN2003 IN2 -> Stepper Coil 2
GPIO 2 -> ULN2003 IN3 -> Stepper Coil 3
GPIO 1 -> ULN2003 IN4 -> Stepper Coil 4
// ---------- Servo Actuators ----------
GPIO 21 -> Servo 1 Signal -> Pan Servo PWM
GPIO 47 -> Servo 2 Signal -> Tilt Servo PWM
GPIO 48 -> Servo 3 Signal -> Dump Servo PWM
// ---------- I2C Color Sensor ----------
GPIO 38 -> I2C SDA -> Color Sensor SDA
GPIO 39 -> I2C SCL -> Color Sensor SCL
+5V_STEP_MOTORS
TILT
M4
SG90
DUMP
M5
SG90
PAN
M1
SG90
SERVO MOTORS
+5V_STEP_MOTORS
BYJ48
J1
STEPPER MOTOR
U8
MPU6050
MPU6050
U5
RGB SENSOR
TCS7345
3V3
U1
JGA25-370
U6
JGA25-370
U12
JGA25-370
U7
JGA25-370
U3
DRV8833
U4
DRV8833
C6
100nF
C5
100nF
DC MOTORS
TESTSUZAKURA KNIGHTS
MADE BY OMOSHSAX
Sheet: /
File: ROBOTICS_DOJO.kicad_sch
Title:
Size: A4
Date: 2026-09-14
Rev: V1.0
KiCad E.D.A. 10.0.6
Id: 1/1